Foundations of Expectation maximization algorithm

[Expectation maximization algorithm]

1.0 Content Block

τ j (t) f (x i ; μ j (t) , Σ j (t)) τ 1 (t) f (x i ; μ 1 (t) , Σ 1 (t)) + τ 2 (t) f (x i ; μ 2 (t) , Σ 2 (t)) . {\displaystyle {\begin{aligned}T_{j,i}^{(t)}:={}&\operatorname {P} (Z_{i}=j\mid X_{i}=\mathbf {x} _{i};\theta ^{(t)})\\={}&{\frac {\tau _{j}^{(t)}\,f{\left(\mathbf {x} _{i};{\boldsymbol {\mu }}_{j}^{(t)},\Sigma _{j}^{(t)}\right)}}{\tau _{1}^{(t)}\,f{\left(\mathbf {x} _{i};{\boldsymbol {\mu }}_{1}^{(t)},\Sigma _{1}^{(t)}\right)}+\tau _{2}^{(t)}\,f{\left(\mathbf {x} _{i};{\boldsymbol {\mu }}_{2}^{(t)},\Sigma _{2}^{(t)}\right)}}}.\end{aligned}}}

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Alternatives EM typically converges to a local optimum, not necessarily the global optimum, with no bound on the convergence rate in general. It is possible that it can be arbitrarily poor in high dimensions and there can be an exponential number of local optima. Hence, a need exists for alternative methods for guaranteed learning, especially in the high-dimensional setting. Alternatives to EM exist with better guarantees for consistency, which are termed moment-based approaches or the so-called spectral techniques. Moment-based approaches to learning the parameters of a probabilistic model enjoy guarantees such as global convergence under certain conditions unlike EM which is often plagued by the issue of getting stuck in local optima. Algorithms with guarantees for learning can be derived for a number of important models such as mixture models, HMMs etc. For these spectral methods, no spurious local optima occur, and the true parameters can be consistently estimated under some regularity conditions.

3.0 Content Block

Proof of correctness Expectation-Maximization works to improve Q (θ ■ θ (t)) {\displaystyle Q({\boldsymbol {\theta }}\mid {\boldsymbol {\theta }}^{(t)})} rather than directly improving log ■ p (X ■ θ) {\displaystyle \log p(\mathbf {X} \mid {\boldsymbol {\theta }})} . Here it is shown that improvements to the former imply improvements to the latter. For any Z {\displaystyle \mathbf {Z} } with non-zero probability ■ p (Z ■ X , θ) {\displaystyle p(\mathbf {Z} \mid \mathbf {X} ,{\boldsymbol {\theta }})} ■, we can write

4.0 Content Block

Further reading Hogg, Robert; McKean, Joseph; Craig, Allen (2005). Introduction to Mathematical Statistics. Upper Saddle River, NJ: Pearson Prentice Hall. pp. 359–364. Dellaert, Frank (February 2002). The Expectation Maximization Algorithm (PDF) (Technical Report number GIT-GVU-02-20). Georgia Tech College of Computing. gives an easier explanation of EM algorithm as to lowerbound maximization. Bishop, Christopher M. (2006). Pattern Recognition and Machine Learning. Springer. ISBN 978-0-387-31073-2. Gupta, M. R.; Chen, Y. (2010). "Theory and Use of the EM Algorithm". Foundations and Trends in Signal Processing. 4 (3): 223–296. CiteSeerX 10.1.1.219.6830. doi:10.1561/2000000034. A well-written short book on EM, including detailed derivation of EM for GMMs, HMMs, and Dirichlet. Bilmes, Jeff (1997). A Gentle Tutorial of the EM Algorithm and its Application to Parameter Estimation for Gaussian Mixture and Hidden Markov Models (Technical Report TR-97-021). International Computer Science Institute. includes a simplified derivation of the EM equations for Gaussian Mixtures and Gaussian Mixture Hidden Markov Models. McLachlan, Geoffrey J.; Krishnan, Thriyambakam (2008). The EM Algorithm and Extensions (2nd ed.). Hoboken: Wiley. ISBN 978-0-471-20170-0.

5.0 Content Block

However, Gibbs' inequality tells us that H (θ ■ θ (t)) ≥ H (θ (t) ■ θ (t)) {\displaystyle H({\boldsymbol {\theta }}\mid {\boldsymbol {\theta }}^{(t)})\geq H({\boldsymbol {\theta }}^{(t)}\mid {\boldsymbol {\theta }}^{(t)})} , so we can conclude that

6.0 Content Block

E step Given our current estimate of the parameters θ(t), the conditional distribution of the Zi is determined by Bayes' theorem to be the proportional height of the normal density weighted by τ:

7.0 Content Block

σ ^ w 2 = 1 N ∑ k = 1 N (x ^ k + 1 − F ^ x ^ k) 2 , {\displaystyle {\widehat {\sigma }}_{w}^{2}={\frac {1}{N}}\sum _{k=1}^{N}{({\widehat {x}}_{k+1}-{\widehat {F}}{\widehat {x}}_{k})}^{2},}

8.0 Content Block

log ■ p (X ■ θ) = log ■ p (X , Z ■ θ) − log ■ p (Z ■ X , θ) . {\displaystyle \log p(\mathbf {X} \mid {\boldsymbol {\theta }})=\log p(\mathbf {X} ,\mathbf {Z} \mid {\boldsymbol {\theta }})-\log p(\mathbf {Z} \mid \mathbf {X} ,{\boldsymbol {\theta }}).}

9.0 Content Block

Maximization step: Choose θ {\displaystyle \theta } to maximize ■ F {\displaystyle F} ■: θ (t + 1) = arg ■ max θ ■ F (q (t) , θ) {\displaystyle \theta ^{(t+1)}=\mathop {\arg \max } _{\theta }F{\left(q^{(t)},\theta \right)}}

10.0 Content Block

History The EM algorithm was explained and given its name in a classic 1977 paper by Arthur Dempster, Nan Laird, and Donald Rubin. They pointed out that the method had been "proposed many times in special circumstances" by earlier authors. One of the earliest is the gene-counting method for estimating allele frequencies by Cedric Smith. Another was proposed by H.O. Hartley in 1958, and Hartley and Hocking in 1977, from which many of the ideas in the Dempster–Laird–Rubin paper originated. Another one by S.K Ng, Thriyambakam Krishnan and G.J McLachlan in 1977. Hartley's ideas can be broadened to any grouped discrete distribution. A very detailed treatment of the EM method for exponential families was published by Rolf Sundberg in his thesis and several papers, following his collaboration with Per Martin-Löf and Anders Martin-Löf. The Dempster–Laird–Rubin paper in 1977 generalized the method and sketched a convergence analysis for a wider class of problems. The Dempster–Laird–Rubin paper established the EM method as an important tool of statistical analysis. See also Meng and van Dyk (1997). The convergence analysis of the Dempster–Laird–Rubin algorithm was flawed and a correct convergence analysis was published by C. F. Jeff Wu in 1983. Wu's proof established the EM method's convergence also outside of the exponential family, as claimed by Dempster–Laird–Rubin.

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F ^ = ∑ k = 1 N (x ^ k + 1 − F ^ x ^ k) 2 ∑ k = 1 N x ^ k 2 . {\displaystyle {\widehat {F}}={\frac {\sum \limits _{k=1}^{N}{\left({\widehat {x}}_{k+1}-{\widehat {F}}{\widehat {x}}_{k}\right)}^{2}}{\sum \limits _{k=1}^{N}{\widehat {x}}_{k}^{2}}}.}

12.0 Content Block

Properties Although an EM iteration does increase the observed data (i.e., marginal) likelihood function, no guarantee exists that the sequence converges to a maximum likelihood estimator. For multimodal distributions, this means that an EM algorithm may converge to a local maximum of the observed data likelihood function, depending on starting values. A variety of heuristic or metaheuristic approaches exist to escape a local maximum, such as random-restart hill climbing (starting with several different random initial estimates ■ θ (t) {\displaystyle {\boldsymbol {\theta }}^{(t)}} ■), or applying simulated annealing methods. EM is especially useful when the likelihood is an exponential family, see Sundberg (2019, Ch. 8) for a comprehensive treatment: the E step becomes the sum of expectations of sufficient statistics, and the M step involves maximizing a linear function. In such a case, it is usually possible to derive closed-form expression updates for each step, using the Sundberg formula (proved and published by Rolf Sundberg, based on unpublished results of Per Martin-Löf and Anders Martin-Löf). The EM method was modified to compute maximum a posteriori (MAP) estimates for Bayesian inference in the original paper by Dempster, Laird, and Rubin. Other methods exist to find maximum likelihood estimates, such as gradient descent, conjugate gradient, or variants of the Gauss–Newton algorithm. Unlike EM, such methods typically require the evaluation of first and/or second derivatives of the likelihood function.